
SpendLog AI

Capability Report

Feature Inventory & Functional Capabilities

Document No: DOC-02

Project: Agentic Expense Tracker (SpendLog AI)

Repository: <https://github.com/Harish-S28/Agentic-expense-tracker>

Prepared for: Project Documentation / README reference

1. Purpose

This report enumerates every functional capability implemented in SpendLog AI, grouped by domain, with the concrete backend route or module responsible for each. It is intended as a capability checklist for contributors, reviewers, and resume/portfolio write-ups.

2. Expense Management

2.1 Core CRUD

Capability	Description	Implementation
Add expense	Log an expense with amount, category, note, and date.	POST /api/expenses
List / filter expenses	Filter by category, month, exact date, or free-text search across note/category.	GET /api/expenses
Delete expense	Remove a single expense by ID.	DELETE /api/expenses/<id>
List distinct categories	Returns all categories currently in use, for filtering.	GET /api/categories

3. Analytics & Reporting

3.1 Analytics engine

Capability	Description	Implementation
Aggregate totals	Sum & count of expenses, optionally scoped by month.	GET /api/analytics
Category breakdown	Spending grouped by category, sorted descending by total.	GET /api/analytics (by_category)
Top spending days	Top 10 highest-spend individual days.	GET /api/analytics (by_date)
6-month trend	Rolling six-month total-spend trend line.	GET /api/analytics (trend)
Monthly AI report	Full monthly review: totals, top day, month-over-month change.	GET /api/data/ai/summary/month
Yearly AI report	Annual review: totals, best/worst month, category breakdown, savings potential.	GET /api/data/ai/summary/year

4. Budgeting

4.1 Dynamic daily-limit budgeting

Capability	Description	Implementation
Set monthly budget	User enters a target monthly budget (e.g. \$500).	POST /api/budget
Live budget status	Base daily limit, cumulative carry-over, effective limit, over/under status.	GET /api/budget/status
Carry-over engine	Unspent daily allowance rolls forward as a bonus; overspending rolls forward as a penalty.	GET /api/budget/rollforward
AI budget tip	Personalized 1-2 sentence coaching message based on budget status.	GET /api/budget/tip

5. Personalization & Onboarding

5.1 Persona-driven experience

Capability	Description	Implementation
User profile	Name, profession, and income stored in a GET/POST API profile.	GET/POST API profile.
5 built-in personas	Student, Doctor, Business Person, Parent, Employee, Non-employee. Each with tailored priorities & tips.	PROFESSOR, NON_EMPLOYEE, EMPLOYEE, PARENT, STUDENT.
Persona-aware AI output	Every AI response (suggestion, tip, chat, analysis) is contextualized on the active persona.	AI uses instructions on the active persona.

6. Agentic AI Features

6.1 AI capabilities

Capability	Description	Implementation
Per-expense AI suggestion	Contextual acknowledgement + question + PROBE/spin/suggest for each expense.	PROBE/spin/suggest.
AI chatbot	Free-form chat, preloaded with 30-day spend history. Supports at least 10 turns of history.	POST/chat/initial.
Deep monthly/yearly analysis	200-250 word narrative report with category-specific analysis.	GET/tip/analysis.
Automatic model selection	Probes gemini-2.5-flash / 2.0-flash / 1.5-flash. AI flash-latest and locks the first that responds.	AI flash-latest and.
Rule-based fallback engine	Deterministic responses using a category/profession/methods fallback agent. Gemini is unavailable.	fallback methods fallback agent.

7. Reminders & Notifications

7.1 Engagement features

Capability	Description	Implementation
Twice-daily reminder check	Backend reports whether any expense has been tagged to check.	GET/tagged to check.
Browser push notification	Native desktop/mobile notification fired in app 9:00-9:16 AM and 7:00-7:16 PM (Merged notifications).	app 9:00-9:16 AM and 7:00-7:16 PM (Merged notifications).
60-second polling loop	Lightweight client-side interval checks the app is not stale (checkRemindersTimes, 60000).	app is not stale (checkRemindersTimes, 60000).

8. Feature Summary Matrix

Domain	# Capabilities	Works without Gemini key?
Expense CRUD	4	Yes — fully independent of AI
Analytics	6	Yes — AI reports fall back to templated summaries
Budgeting	4	Yes — AI tip falls back to rule-based tip
Personalization	3	Yes — persona data is separate from AI
Agentic AI	5	Partially — degrades to deterministic responses

Reminders	3	Yes — entirely rule-based, no AI involved
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Overall, 25 discrete capabilities span 6 functional domains. A defining trait of the system is that every AI-touched capability remains fully operational — with reduced personalization — even with zero external API dependency, which is a strong resume talking point: *'designed for graceful degradation with 100% feature availability offline.'*